Mohamed Elmugtaba Mohamed

BIOINFORMATICS PRACTITIONER | GENOMIC DATA ANALYSIS & REPRODUCIBLE PIPELINES | PHARMACIST

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PROFESSIONAL SUMMARY

> *A pharmacist who retrained in computational biology and builds reproducible, tested genomic workflows.*

Pharmacist with 7+ years across pharmacy practice, GMP pharmaceutical manufacturing and medical and scientific sales. Builds Nextflow DSL2 pipelines on Linux for germline variant calling (GATK4, joint genotyping) and RNA-seq, with Git, modular code, nf-test automated testing, GitHub Actions CI and Docker containers, and has hands-on single-cell RNA-seq analysis in Python (Scanpy). GMP production work (batch-record approval, SOPs, deviation records, 100% compliance on every product release) built the documentation and quality habits that reproducible analysis depends on; scientific sales built clear communication of evidence to physicians. Seeking an entry-level bioinformatics role in genomics or precision medicine.

BIOINFORMATICS PROJECTS

1. Germline Variant Calling & Joint Genotyping Pipeline (BAM → VCF)

↗ GitHub

`Nextflow DSL2` `GATK4` `Samtools` `bcftools` `MultiQC` `nf-test` `GitHub Actions` `Docker`

> **Output:** Per-sample GVCFs and one joint-genotyped cohort VCF of germline SNPs and indels, with alignment and variant QC in a MultiQC report; nf-test suite and CI passing.

SCALE	Family trio (3 samples) · chromosome 20 slice · hg19/b37 · starts from aligned BAMs.
METHODS	Samtools index → HaplotypeCaller (GVCF) → GenomicsDBImport → GenotypeGVCFs; samtools stats/flagstat and bcftools stats → MultiQC. Modular layout, nf-schema parameter validation, fail-fast input checks, automatic retry on out-of-memory exits.
SOURCE	Dataset and core commands adapted from Seqera's Nextflow for Science: Genomics course (credited in the repo).

2. Reproducible Human RNA-seq Pipeline (FASTQ → counts)

↗ GitHub

`Nextflow DSL2` `FastQC` `fastp` `STAR` `featureCounts` `MultiQC` `Docker`

> **Output:** Raw FASTQ → sorted BAM → gene-level count matrix, with all QC aggregated in one MultiQC report.

SCALE	Human chromosome 22 (GRCh38, Ensembl 110) · single- or paired-end reads.
METHODS	STAR index → FastQC → fastp → STAR → featureCounts → MultiQC; Docker containers on a local executor (containerised, not yet run on HPC or cloud); fail-fast checks on the samplesheet and references.

3. COVID-19 vs Control PBMC Single-Cell RNA-seq Analysis

↗ GitHub

`Python` `Scanpy` `AnnData` `Scrublet` `Scanorama` `gseapy`

> **Output:** QC-filtered 9,000 cells to 5,589; compared three batch-integration methods; clustered (Louvain, 9 clusters), annotated by reference label transfer and Enrichr markers, and compared COVID-19 with control cells within clusters.

SCALE	6 samples (3 COVID-19, 3 controls) · 10x Genomics · 1,500 cells each.
METHODS	Mito/ribo QC, sex and cell-cycle checks, doublet removal; MNN, ComBat, Scanorama; Wilcoxon, t-test and logistic-regression DE; Enrichr and GSEA.
SOURCE	NBIS project-based scRNA-seq workshop.

4. Single-Cell & Spatial Analyses TUTORIAL-BASED

Visium + scRNA-seq integration: human lymph node QC, clustering and spatially variable genes (SpatialDE); two mouse-brain sections integrated with Scanorama and labelled from a cortex reference. ↗ GitHub

Scarf: 10x 5K PBMC clustering, and projection of 14K interferon-stimulated PBMCs onto a 15K control reference with label transfer. ↗ GitHub

TECHNICAL SKILLS

LANGUAGES & ENVIRONMENT

`Python` `R (ggplot2)` `Bash` `Linux / Unix` `Jupyter`

WORKFLOWS & PRACTICES

`Nextflow DSL2` `nf-test` `nf-schema` `nf-core*` `Docker` `Conda` `Git / GitHub` `GitHub Actions CI`

NGS, VARIANT CALLING & QC

`GATK4` `HaplotypeCaller` `GenomicsDBImport` `GenotypeGVCFs` `Samtools` `bcftools` `STAR` `featureCounts` `FastQC` `fastp` `MultiQC`

SINGLE-CELL & SPATIAL

`Scanpy` `AnnData` `Scanorama` `ComBat / MNN` `Scrublet` `gseapy` `Scarf*` `SpatialDE*`

* course- or tutorial-level

EDUCATION

BSc Pharmacy

National Ribat University

Khartoum, Sudan · 2010 – 2015

Coursework: biochemistry, microbiology, pharmacology, clinical pharmacology, analytical chemistry.

CERTIFICATIONS

PIPELINES & NGS

- **Nextflow Training Week 2026-Q1: 5 certificates**
  Hello Nextflow · Nextflow Run · Hello nf-core · Nextflow for RNAseq · Nextflow for Genomics
  Seqera · Feb 2026

BIOINFORMATICS

- **Bioinformatics for Biologists: An Introduction to Linux, Bash Scripting, and R**
  Wellcome Connecting Science / FutureLearn · Jul 2023 · 100%
- **scRNA-seq workshop (project-based)**
  NBIS, National Bioinformatics Infrastructure Sweden
- **Python**
  Kaggle · Feb 2022

TRAINING & TRANSITION

Self-directed retraining in computational biology

2022 – 2024

Career break after a family relocation from Sudan during regional instability, used for structured self-study.

- ✓ **Foundations:** Python, Kaggle (Feb 2022)
- ✓ **Portfolio:** single-cell and spatial analyses on GitHub (Jul–Sep 2022), including the NBIS COVID-19 workshop project
- ✓ **Visualisation:** R and ggplot2 scripts (Jul 2023)
- ✓ **Coursework:** Linux, Bash scripting and R for biologists, Wellcome Connecting Science, 100% (Jul 2023)

Pipeline engineering

2025 – Present

Relocated to Riyadh (Oct 2025); structured training and pipeline building alongside the job search.

- ✓ **Seqera (Feb 2026):** 5 Nextflow certificates, including RNA-seq, genomics and nf-core
- ✓ **Built:** RNA-seq pipeline (Feb 2026) and GATK4 germline variant-calling pipeline with nf-test and CI (Sep–Oct 2026)
- ✓ **Cloud:** Google Cloud Digital Leader badge, Google Skills (Jan 2026)
- ✓ **In progress:** Integrating Generative AI into Data Workflows (Google Skills)

PROFESSIONAL EXPERIENCE

Key Account & Operations Coordinator

2024 – 2025

Dukhan Group · Qatar

- Built and maintained operational trackers (contract status, SLA compliance, recovery outcomes), checking the data across business units.
- Found the root causes of workflow bottlenecks between departments; primary liaison between clients and internal teams to resolve service failures.

Scientific Sales Officer

2021 – 2022

Salmawit Co. Ltd · Sudan

- Turned published clinical study data into evidence-based communication for physicians on efficacy, safety and appropriate prescribing across a GIT/IBS portfolio.
- Planned and delivered CME events and clinical seminars in gastroenterology at least once a quarter; ran a structured call cycle of 120 physicians.

Production Pharmacist

2019 – 2020

Blue Nile Pharmaceutical Factory · Sudan

- Reviewed and approved batch production and manufacturing records (BPR/BMR) before release, with 100% GMP compliance on every product release.
- Wrote SOPs and kept deviation reports and traceability logs.
- Supervised end-to-end production and a 90-person team, coordinating QA, QC and maintenance as monthly output roughly quadrupled (15–17M → 60–65M tablets/month).

Medical Representative (Territory Lead)

2017 – 2018

Aurobindo Pharma · Sudan

- Led the launch of a new antibiotic brand in a high-priority territory and secured a government formulary listing under the national government insurance scheme; gave scientific presentations and roundtables for key opinion leaders.

Medical Representative

2016 – 2017

Bioderma · Sudan

Pharmacist

2015 – 2016

Wenji Pharmacy & Sudan Military Hospital · Sudan

- Dispensing, dose verification and drug-interaction screening; patient counselling on adherence and side effects.

LANGUAGES Arabic (native) · English (full professional proficiency) · **STATUS** Valid, transferable Iqama (KSA) · **AVAILABILITY** Immediate

Updated October 2026